

Victor Sechaud

PHD STUDENT · APPLY MATHEMATICS



About me

I am a PhD student in Applied Mathematics at ENS de Lyon, supervised by Julián Tachella and Patrice Abry. I'm working on inverse problems and machine learning, in particular, self-supervised learning methods for non-linear problems such as phase retrieval and declipping.

Interests

- Machine learning
- Deep learning
- Optimization
- Statistics
- Compressive sensing
- Zero-order optimization

Contact

✉ victor.sechaud@ens-lyon.fr

Personal

Nationality: French
Date of birth: 15.11.1999

Languages

French	Native
English	B2+

[vsechaud.github.io/](https://github.com/vsechaud)
<https://github.com/vsechaud>

EDUCATION

- | | | |
|----------|---|--------------|
| 2023-now | PhD in Apply Mathematics
CNRS, ENS DE LYON,
Thesis: "Self-supervised learning for non-linear inverse problems" | LYON, FRANCE |
| 2023 | Master's degree
UNIVERSITY LYON 1 CLAUDE BERNARD
Apply mathematics | LYON, FRANCE |
| 2022 | Master's degree
UNIVERSITY LYON 1 CLAUDE BERNARD
Agrégation de mathématiques (Highly selective French national teaching qualification) | LYON, FRANCE |
| 202 | Undergraduate degree
UNIVERSITY LYON 1 CLAUDE BERNARD
Mathematics | LYON, FRANCE |

ACADEMIC EXPERIENCE

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| 2025 sept-now | Visiting scholar
HERIOT-WATT UNIVERSITY
9-week visiting researcher with Marcelo Pereyra | EDINBURGH, UK |
| 2023-now | Teaching
ENS DE LYON
Mathematics department: Linear Algebra, Python, and Statistics | LYON, FRANCE |

PUBLICATIONS

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|------|---|--|
| 2026 | V. Sechaud , L. Jacques, P. Abry, and J. Tachella, "Self-supervised learning for phase retrieval: theoretical analysis and experimental results", <i>Journal paper to be submitted soon</i> , 2026.
V. Sechaud , L. Jacques, P. Abry, and J. Tachella, "Learning to reconstruct from saturated data: audio declipping and high-dynamic range imaging", <i>TMLR</i> , 2026.
V. Sechaud , J. Scanvic, Q. Barthélemy, P. Abry, and J. Tachella, "Equivariant Splitting: Self-supervised learning from incomplete data", <i>ICLR</i> , 2026. | |
| 2025 | J. Tachella, L. Davy, J. Scanvic, V. Sechaud , R. Vo, et al, "DeepInverse: A Python package for solving imaging inverse problems with deep learning", <i>Journal of Open Source Software</i> , 2025.
V. Sechaud , P. Abry, L. Jacques, and J. Tachella, "Apprentissage auto-supervisé pour la reconstruction de phase", <i>GRETSI</i> , 2025. | |
| 2024 | V. Sechaud , L. Jacques, P. Abry, and J. Tachella, "Equivariance-based self-supervised learning for audio signal recovery from clipped measurements", <i>EUSIPCO 2024</i> , 2024. | |

TALKS

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|------|--|------------------------|
| 2026 | Equivariant splitting: self-supervised learning method
ICLR (POSTER) | RIO DE JANEIRO, BRAZIL |
| 2026 | Reconstruct Anything Model (RAM)
ICLR (POSTER) | RIO DE JANEIRO, BRAZIL |
| 2025 | Combining equivariance with splitting methods
HERIOT-WATT UNIVERSITY | EDINBURGH, UK |
| 2025 | Self-supervised learning for phase retrieval problem
GRETSI | STRASBOURG, FRANCE |
| 2024 | learning to reconstruct from clipped signals alone
EUSIPCO | LYON, FRANCE |